

**SHYAMA PRASAD MUKHERJI
COLLEGE FOR WOMEN
UNIVERSITY OF DELHI
R PROJECT**

**INDIA VS CHINA - ECONOMIC
ANALYSIS**



SUBMITTED BY:

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B.A (Hons.) ECONOMICS

2nd YEAR (3rd SEMESTER)

College Roll No. – 21/1330

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INTRODUCTION

INDIA

India, officially the **Republic of India** (Hindi: *Bhārat Gaṇarājya*) is a country in South Asia. It is the seventh-largest country by area, the second-most populous country, and the most populous democracy in the world. Bounded by the Indian ocean on the south, the Arabian sea on the southwest, and the Bay of Bengal on the southeast, it shares land borders with Pakistan to the west; China, Nepal, and Bhutan to the north; and Bangladesh and Myanmar to the east. In the Indian Ocean, India is in the vicinity of Sri Lanka and the Maldives; its Andaman and Nicobar Islands share a maritime border with Thailand, Myanmar, and Indonesia. The nation's capital city is New Delhi.

CHINA

China, officially the **People's Republic of China (PRC)**, is a country in East Asia. It is the world's most populous country with a population exceeding 1.4 billion people. China spans five geographical time zones and borders fourteen countries by land, the most of any country in the world, tied with Russia. China also has a narrow maritime boundary with the disputed Taiwan. Covering an area of approximately 9.6 million square kilometres (3,700,000 sq. mi), it is the world's third largest country by total land area. The country consists of 23 provinces, five autonomous regions, four municipalities, and two Special Administrative Regions (Hong Kong and Macau). The national capital is Beijing, and the most populous city and financial centre is Shanghai.

AIM OF THIS PROJECT

This project attempts to present the statistical calculations and measurements of the economy of two countries – India and China. Representation of Economical parameters in terms of various methods such as –Histogram, Pie chart, Bar Chart, Boxplot and Line Graph aligned with numbers stored in variables, will provide an inner insight of what these variables have and their correlation with others. The execution of these variables reveals the various facts and figures that are considerable and noticeable. The numerical values connected with each other show marks of significant conditions prevailing in the economy. This plausible analysing becomes into real figures with R.



सत्यमेव जयते



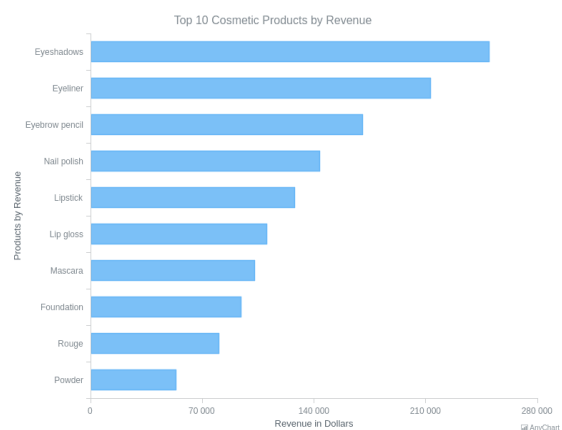
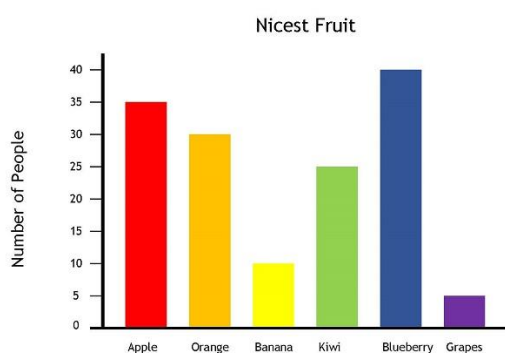
BAR GRAPH

A **bar chart** or **bar graph** is a chart or graph that presents categorical data with rectangular bars with heights or lengths proportional to the values that they represent. The bars can be plotted vertically or horizontally. A vertical bar chart is sometimes called a **column chart**.

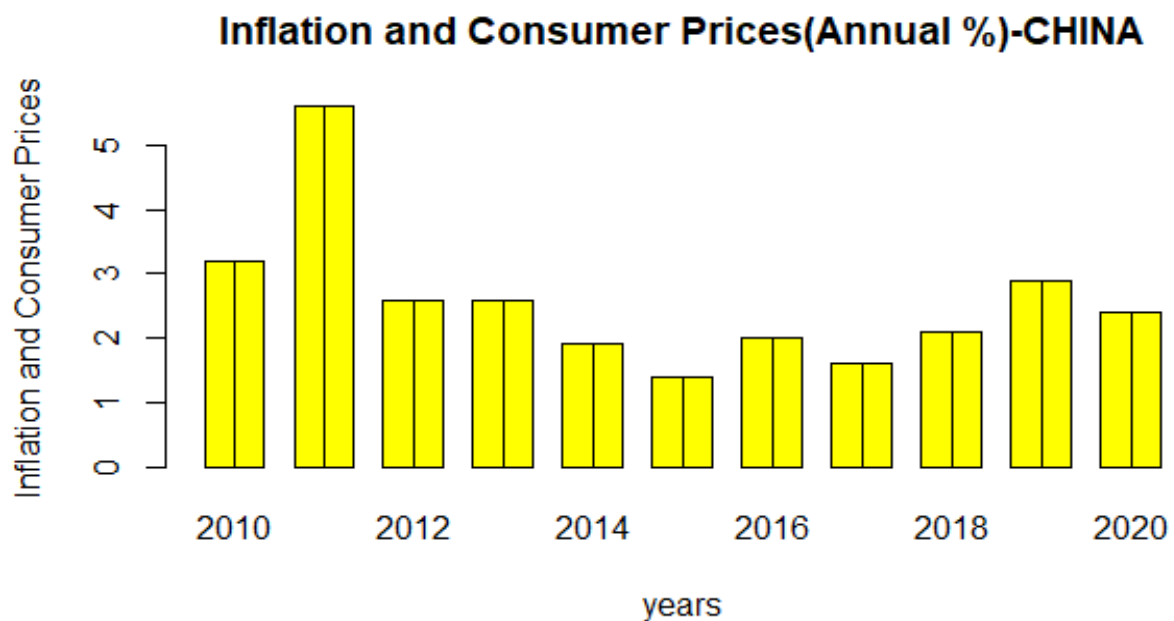
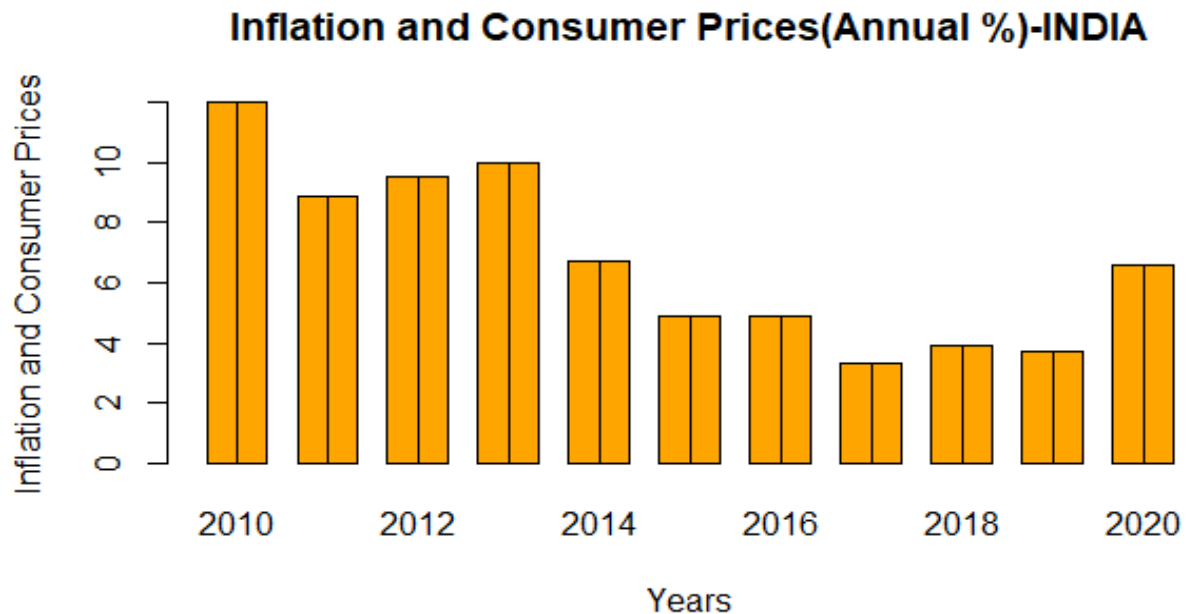
A bar graph shows comparisons among discrete categories. One axis of the chart shows the specific categories being compared, and the other axis represents a measured value. Some bar graphs present bars clustered in groups of more than one, showing the values of more than one measured variable.

Bar graphs/charts provide a visual presentation of categorical data. Categorical data is a grouping of data into discrete groups, such as months of the year, age group, shoe sizes, and animals. These categories are usually qualitative. In a column (vertical) bar chart, categories appear along the horizontal axis and the height of the bar corresponds to the value of each category.

Bar charts have a discrete domain of categories, and are usually scaled so that all the data can fit on the chart. When there is no natural ordering of the categories being compared, bars on the chart may be arranged in any order. Bar charts arranged from highest to lowest incidence are called Pareto charts.



INFLATION ANALYSIS 2010-2020 USING BAR GRAPH



Source: World Bank, Inflation & Consumer Prices (annual %)

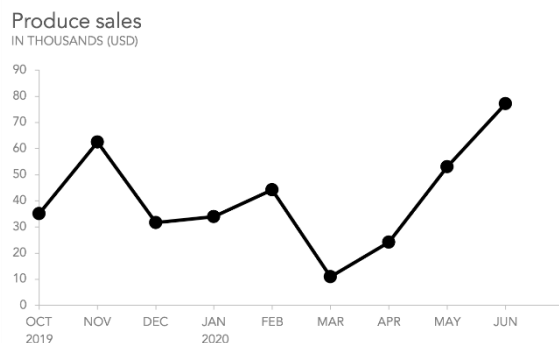
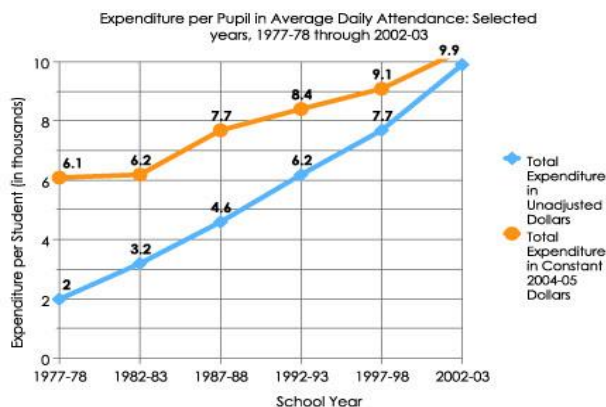
Interpretation: As we can see from the bar graphs, even though India and China are pretty close in GDP and population, inflation in China has been consistently less throughout the years. Whereas, India on the other hand has had a pretty high level of inflation, even double digits for a few years.

LINE GRAPH

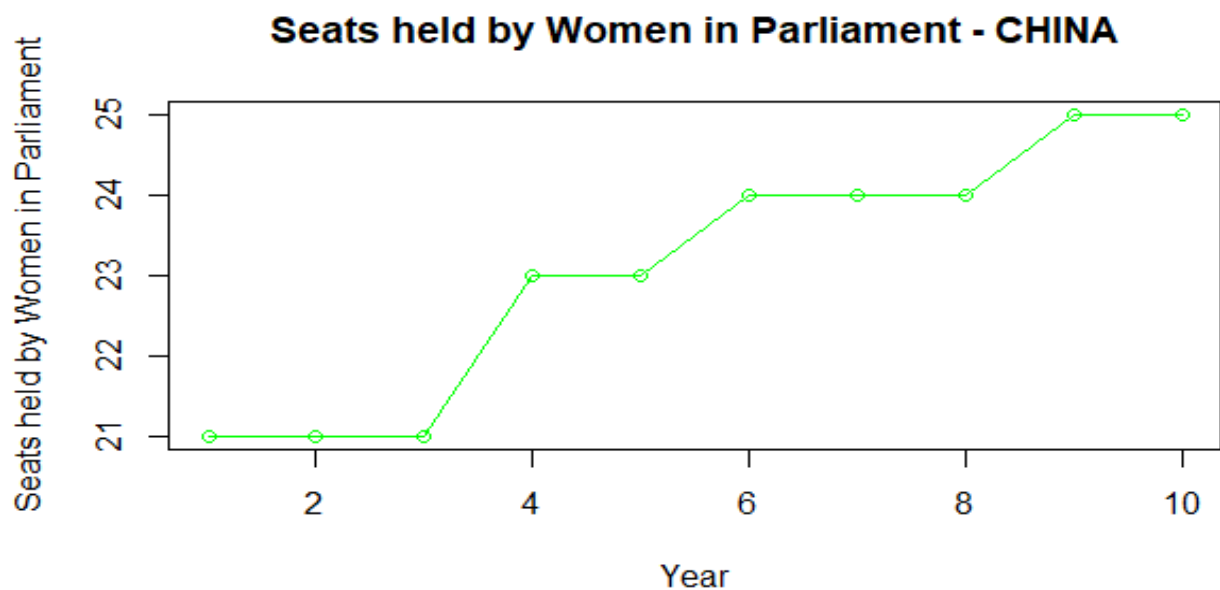
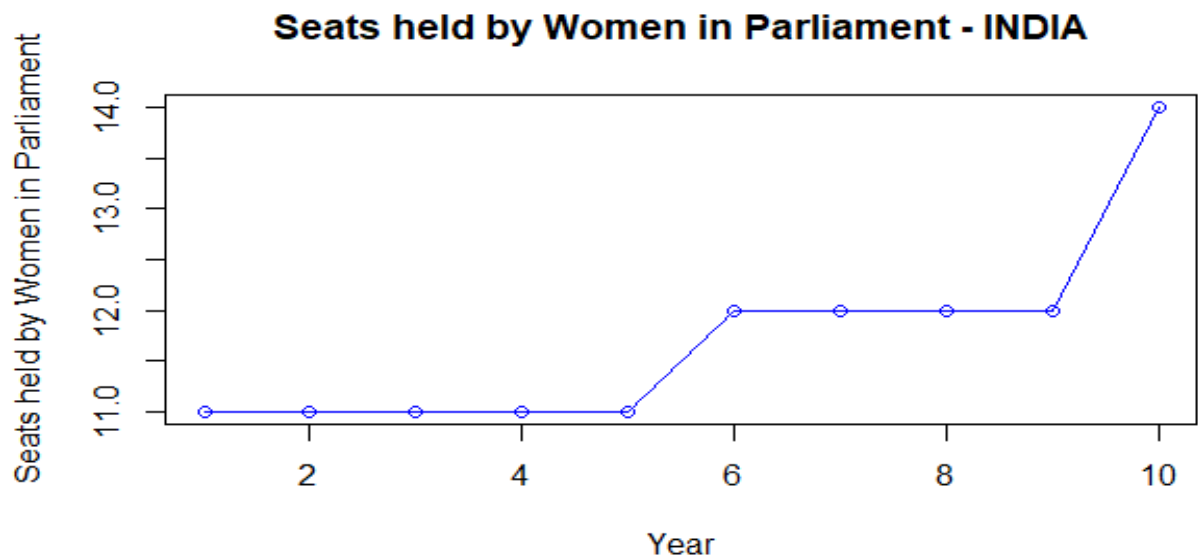
A **line chart** or **line graph** or **curve chart** is a type of chart which displays information as a series of data points called 'markers' connected by straight line segments. It is a basic type of chart common in many fields. It is similar to a scatter plot except that the measurement points are ordered (typically by their x-axis value) and joined with straight line segments. A line chart is often used to visualize a trend in data over intervals of time – a time series – thus the line is often drawn chronologically. In these cases, they are known as run charts.

In the experimental sciences, data collected from experiments are often visualized by a graph. Charts often include an overlaid mathematical function depicting the best-fit trend of the scattered data.

Some of the earliest known line charts are generally credited to Francis Hoqabee, Nicolaus Samuel Cruquius, Johann Heinrich Lambert and William Playfair.



SEATS HELD BY WOMEN IN PARLIAMENT 2010-2020
REPRESENTED BY LINE GRAPH



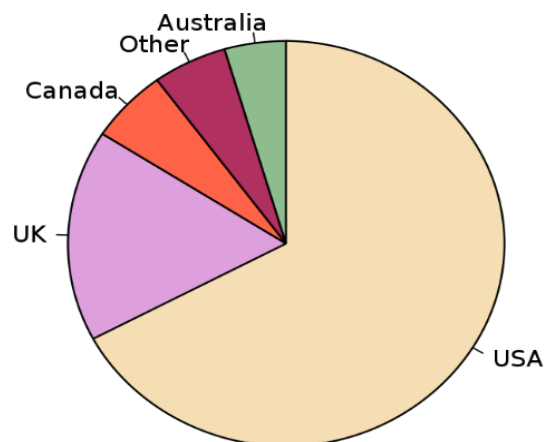
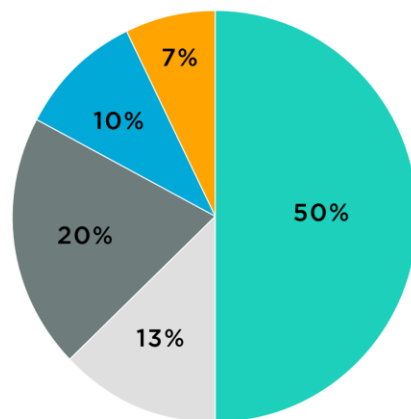
Source: World Bank, Proportion of seats held by women in national parliaments (%)

Interpretation: As we can see from the line graphs, even though India and China are pretty close in GDP and population, participation of women as seats held in national parliaments in China has been consistently higher throughout the years than India.

PIE CHART

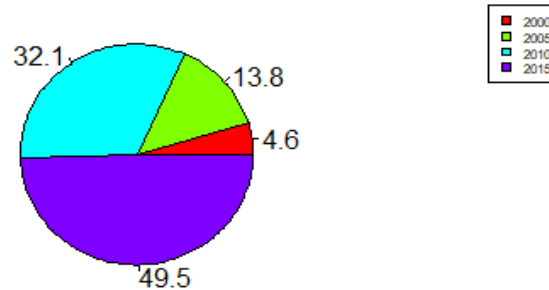
A **pie chart** (or a **circle chart**) is a circular statistical graphic, which is divided into slices to illustrate numerical proportion. In a pie chart, the arc length of each slice (and consequently its central angle and area) is proportional to the quantity it represents. While it is named for its resemblance to a pie which has been sliced, there are variations on the way it can be presented. The earliest known pie chart is generally credited to William Playfair's *Statistical Breviary* of 1801.

Pie charts are very widely used in the business world and the mass media. However, they have been criticized, and many experts recommend avoiding them, as research has shown it is difficult to compare different sections of a given pie chart, or to compare data across different pie charts. Pie charts can be replaced in most cases by other plots such as the bar chart, box plot, dot plot, etc.

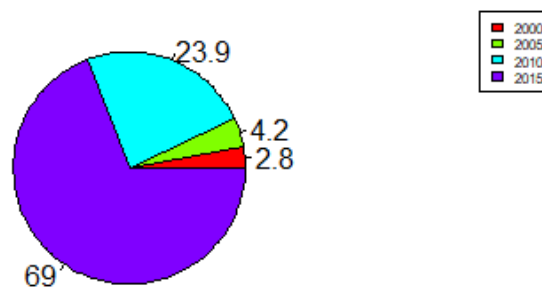


ELECTRICITY PRODUCTION FROM RENEWABLE SOURCES 2000-2015 REPRESENTED BY PIE CHART

Electricity Production from Renewable Sources - INDIA



Electricity Production from Renewable Sources - CHINA



Source: Electricity production from renewable sources, excluding hydroelectricity (% of total)

Interpretation: As we can see from the pie charts, even though India and China are pretty close in GDP and population, electricity production from renewable resources is more in India. Which means India is producing and using more environment friendly energy and less carbon production than China.

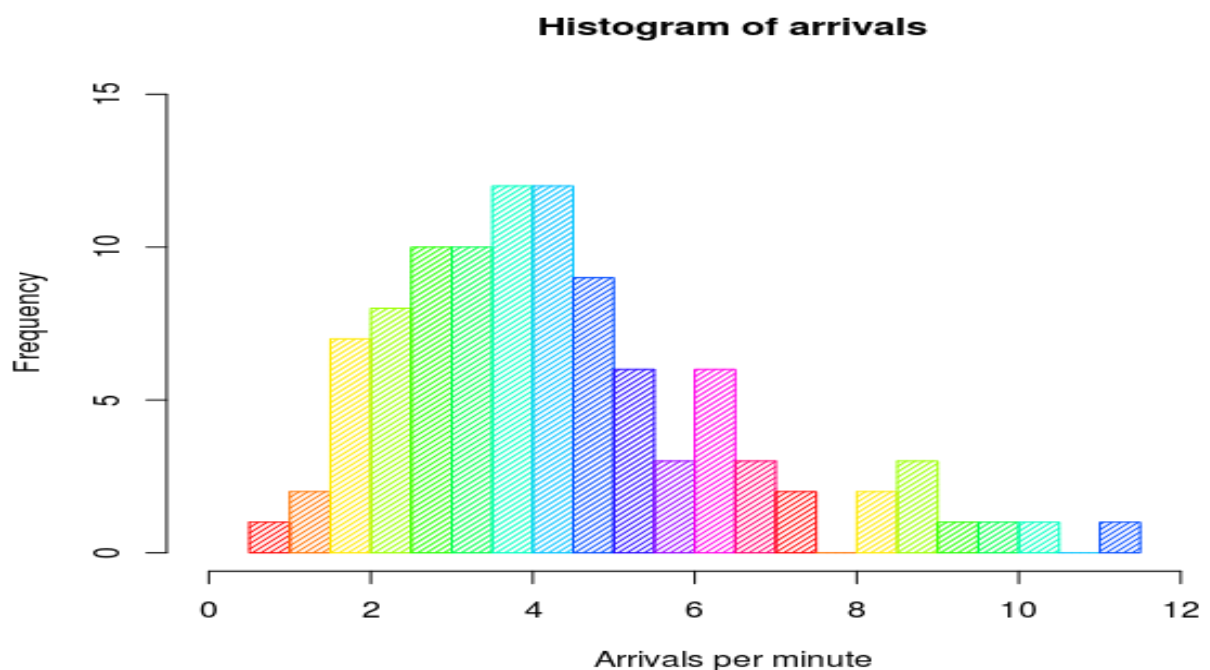
HISTOGRAM

A **histogram** is an approximate representation of the distribution of numerical data. The term was first introduced by Karl Pearson. To construct a histogram, the first step is to "bin" (or "bucket") the range of values—that is, divide the entire range of values into a series of intervals—and then count how many values fall into each interval. The bins are usually specified as consecutive, non-overlapping intervals of a variable. The bins (intervals) must be adjacent and are often (but not required to be) of equal size.

Histograms give a rough sense of the density of the underlying distribution of the data, and often for density estimation: estimating the probability density function of the underlying variable. The total area of a histogram used for probability density is always normalized to 1. If the length of the intervals on the x -axis are all 1, then a histogram is identical to a relative frequency plot.

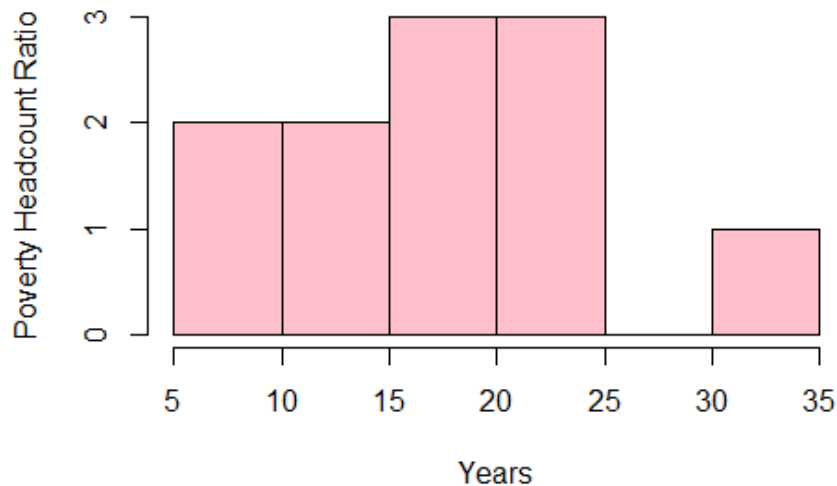
The histogram is one of the seven basic tools of quality control.

Histograms are sometimes confused with bar charts. A histogram is used for continuous data, where the bins represent ranges of data, while a bar chart is a plot of categorical variables. Some authors recommend that bar charts have gaps between the rectangles to clarify the distinction.

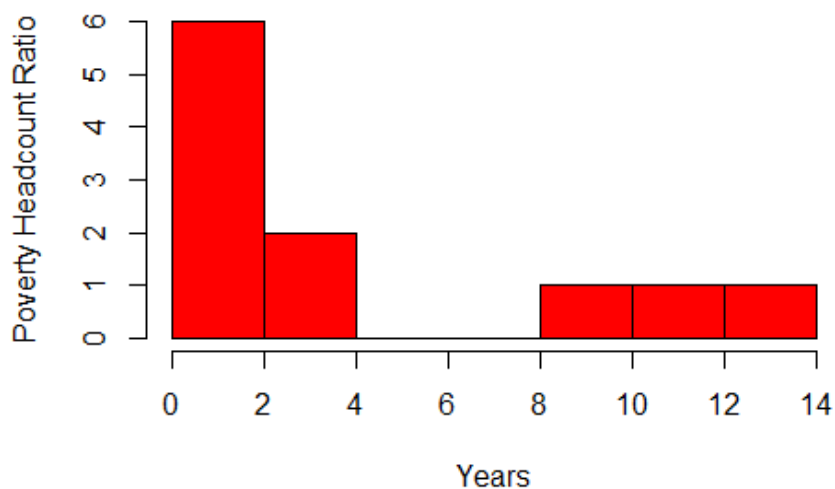


POVERTY HEADCOUNT RATIO 2010-2020 REPRESENTED BY HISTOGRAM

Poverty Headcount Ratio - INDIA



Poverty Headcount Ratio - CHINA



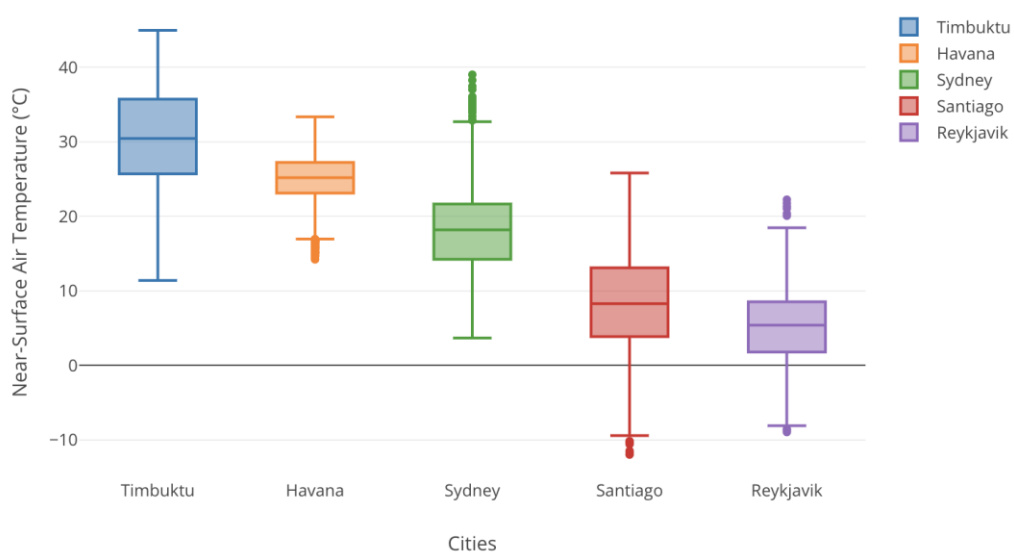
Source: World Bank, Poverty Headcount Ratio at \$2.15 a day (% of population)

Interpretation: As we can see from the histograms, even though India and China are pretty close in GDP and population, poverty headcount ratio in the population for China is lesser than in India. Poverty is declining for both the countries but China records a lesser poverty ratio.

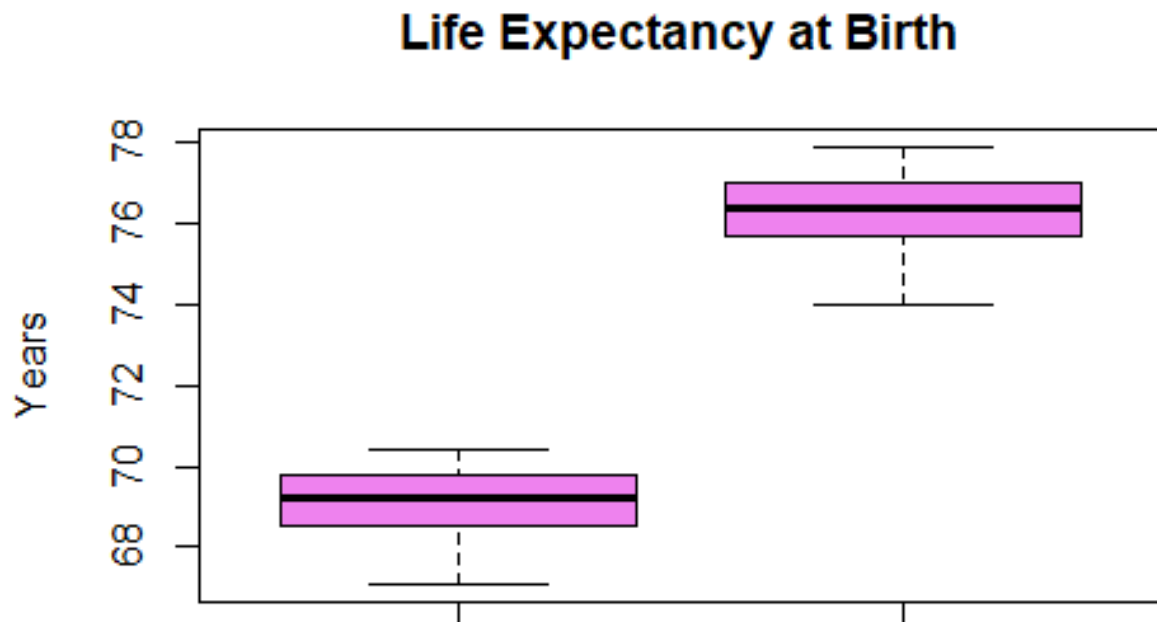
BOX PLOT

In descriptive statistics, a **box plot** or **boxplot** is a method for graphically demonstrating the locality, spread and skewness groups of numerical data through their quartiles. In addition to the box on a box plot, there can be lines (which are called *whiskers*) extending from the box indicating variability outside the upper and lower quartiles, thus, the plot is also termed as the **box-and-whisker plot** and the **box-and-whisker diagram**. Outliers that differ significantly from the rest of the dataset may be plotted as individual points beyond the whiskers on the box-plot. Box plots are non-parametric: they display variation in samples of a statistical population without making any assumptions of the underlying statistical distribution (though Tukey's boxplot assumes symmetry for the whiskers and normality for their length). The spacings in each subsection of the box-plot indicate the degree of dispersion (spread) and skewness of the data, which are usually described using the five-number summary. In addition, the box-plot allows one to visually estimate various L-estimators, notably the interquartile range, midline, range, mid-range, and trimean. Box plots can be drawn either horizontally or vertically.

Box plots



**LIFE EXPECTANCY AT BIRTH 2010-2020 PRESENTED BY
BOX PLOT**



Source: World Bank, Life Expectancy at Birth, total (Years)

Interpretation: Even though India and China have similar living conditions, we can see from the Box Plot that China clearly has a higher level of life expectancy at birth. This gives us an idea that China has better health facilities than India.

GDP REVIEW ON THE BASIS OF DIFFERENT STATISTICAL TOOLS

STATISTICAL TOOLS	INDIA	CHINA
MEAN	2.303333	11.44667
MEDIAN	2.195	11.145
MODE	N/A	N/A
TRIM MEAN	2.279	11.354
LENGTH	12	12
COVARIANCE	0.2358242	11.07993
STANDARD DEVIATION	0.4856174	3.328653
MIN	1.68	6.09
MAX	3.17	17.73
RANGE	1.68 - 3.17	6.09 - 17.73
IQR	0.825	4.6775

Source: World Bank, GDP (Current US \$)

Interpretation: The mean GDP of China is higher than India, China has a mean GDP of 11.44 and that of India is 2.3 as we can see from the table above, on using different statistical tools. Hence, we can interpret that even with similar population sizes China is a much bigger economy than India, it's around 5x times bigger.

CONCLUSION

We know China is 1st in population and India is 2nd population wise, area wise China is the 3rd largest country by land and India is 7th largest country by land, with similar living conditions. This is one of the main reasons why I chose this as a topic – **Economic Analysis between India and China.**

After comparing both the countries on the basis of various graphs like bar graph, line graph, box plot, pie chart, scattered plot and using multiple statistical tools like mean, median, mode, min, max, covariance, correlation and standard deviation we can interpret and deduce some conclusions.

The conclusions are as follows: -

- Inflation in China has been consistently lesser than India, India even saw double digit inflation rates.
- Women in China have always held more seats in the parliament than in India, showing more women representation in China.
- Poverty is declining in both the countries, but Poverty Headcount Ratio in China is lesser than in India even with a similar population size.
- China is larger in size than India, but India produces a larger amount of electricity from renewable resources, shows that India is more environment friendly than China.
- Life Expectancy at Birth in India is lesser than that in China, this gives us the idea that health facilities in China are better than in India.
- The mean GDP of China is higher than that of India, about 5x times bigger as an economy. So, we can say that China is a more prosperous economy in comparison to India.

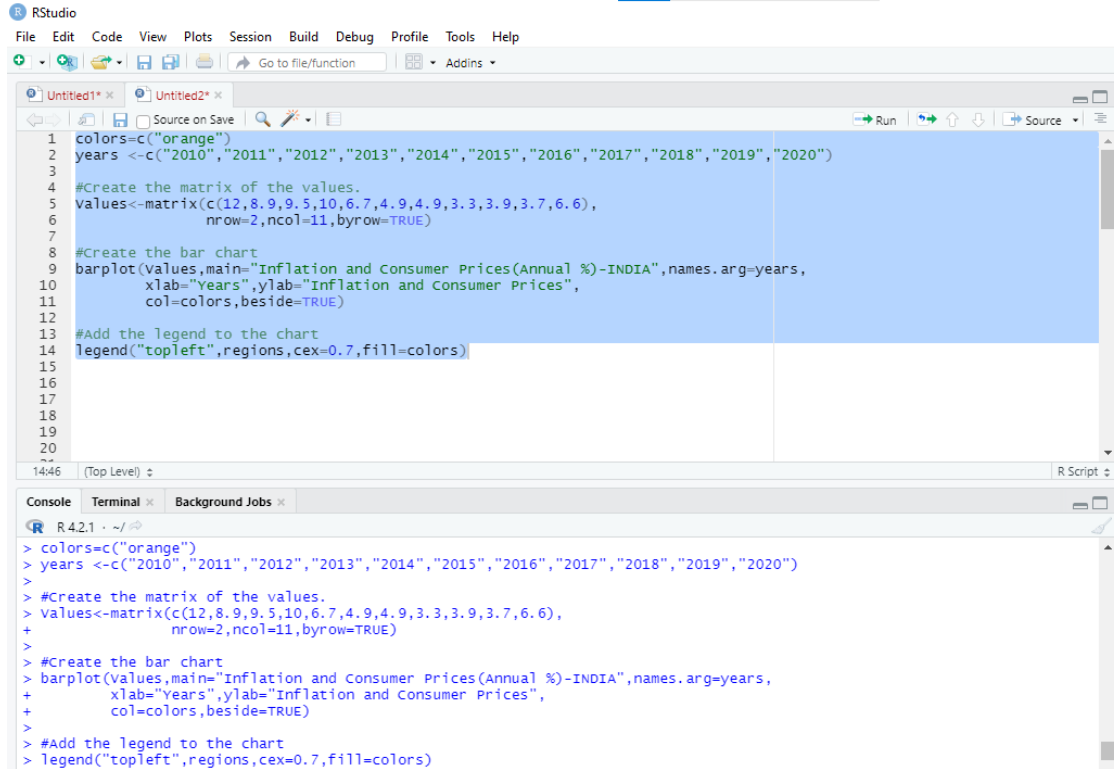
References:

1. <https://data.worldbank.org/>
2. <https://www.wikipedia.org/>

APPENDIX

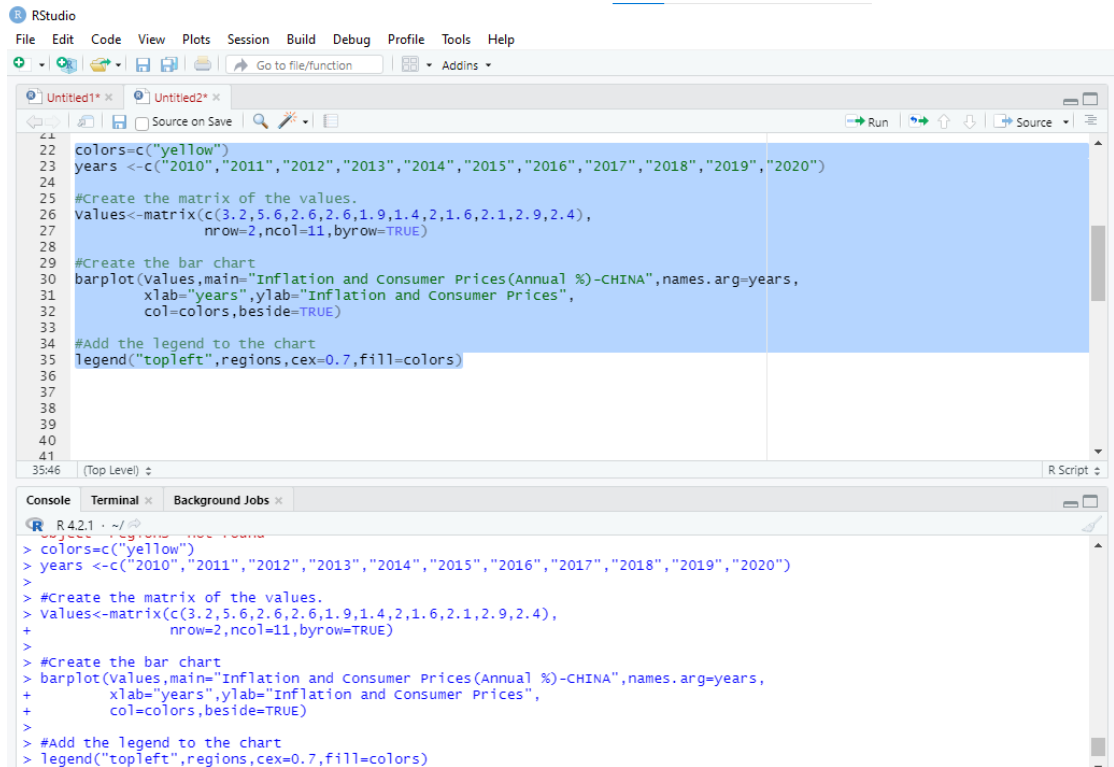
• INFLATION ANALYSIS 2010-2020 USING BAR GRAPH

INDIA



```
1 colors=c("orange")
2 years <-c("2010","2011","2012","2013","2014","2015","2016","2017","2018","2019","2020")
3
4 #Create the matrix of the values.
5 values<-matrix(c(12,8,9,9,5,10,6,7,4,9,3,3,3,9,3,7,6,6),
6               nrow=2,ncol=11,byrow=TRUE)
7
8 #Create the bar chart
9 barplot(values,main="Inflation and Consumer Prices(Annual %)-INDIA",names.arg=years,
10        xlab="Years",ylab="Inflation and Consumer Prices",
11        col=colors,beside=TRUE)
12
13 #Add the legend to the chart
14 legend("topleft",regions,cex=0.7,fill=colors)
```

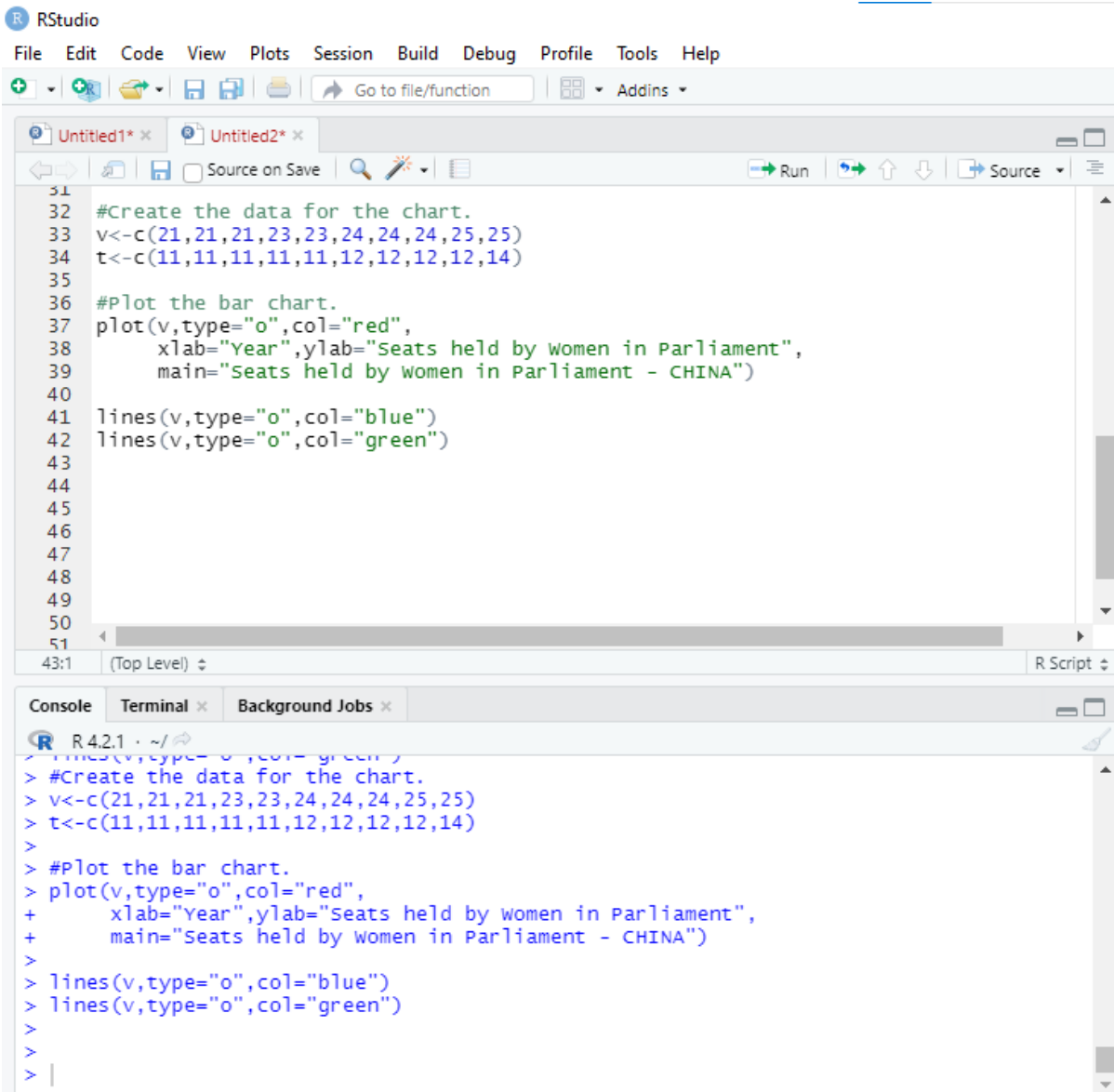
CHINA



```
22 colors=c("yellow")
23 years <-c("2010","2011","2012","2013","2014","2015","2016","2017","2018","2019","2020")
24
25 #Create the matrix of the values.
26 values<-matrix(c(3.2,5.6,2.6,2.6,1.9,1.4,2,1.6,2.1,2.9,2.4),
27               nrow=2,ncol=11,byrow=TRUE)
28
29 #Create the bar chart
30 barplot(values,main="Inflation and Consumer Prices(Annual %)-CHINA",names.arg=years,
31        xlab="years",ylab="Inflation and Consumer Prices",
32        col=colors,beside=TRUE)
33
34 #Add the legend to the chart
35 legend("topleft",regions,cex=0.7,fill=colors)
```


INDIA

CHINA



The screenshot shows the RStudio environment with two untitled files. The script in 'Untitled2*' contains R code to create a bar chart and add lines for China. The console shows the execution of the code.

```
31  
32 #Create the data for the chart.  
33 v<-c(21,21,21,23,23,24,24,24,25,25)  
34 t<-c(11,11,11,11,11,12,12,12,12,14)  
35  
36 #Plot the bar chart.  
37 plot(v,type="o",col="red",  
38       xlab="Year",ylab="Seats held by women in Parliament",  
39       main="Seats held by women in Parliament - CHINA")  
40  
41 lines(v,type="o",col="blue")  
42 lines(v,type="o",col="green")  
43  
44  
45  
46  
47  
48  
49  
50  
51
```

Console output:

```
R 4.2.1: ~/R  
> lines(v,type="o",col="green")  
> #Create the data for the chart.  
> v<-c(21,21,21,23,23,24,24,24,25,25)  
> t<-c(11,11,11,11,11,12,12,12,12,14)  
>  
> #Plot the bar chart.  
> plot(v,type="o",col="red",  
+       xlab="Year",ylab="Seats held by women in Parliament",  
+       main="Seats held by women in Parliament - CHINA")  
>  
> lines(v,type="o",col="blue")  
> lines(v,type="o",col="green")  
>  
>  
>  
> |
```

- **ELECTRICITY PRODUCTION FROM RENEWABLE SOURCES 2000-2015 REPRESENTED BY PIE CHART**

INDIA

The RStudio interface displays the following R code in the editor:

```
44  
45 #Create data for the graph.  
46 geeks<-c(0.5,1.5,3.5,5.4)  
47 labels<-c("2000","2005","2010","2015")  
48  
49 piepercent<-round(100*geeks/sum(geeks),1)  
50  
51 #Plot the chart.  
52 pie(geeks,labels=piepercent,main="Electricity Production from Renewable Sources - INDIA",col=rainbow(length(geeks)))  
53 legend("topright",c("2000","2005","2010","2015"),  
54       cex=0.5,fill=rainbow(length(geeks)))  
55  
56  
57  
58  
59  
60  
61  
62  
63
```

The console shows the execution of the code:

```
> #Create data for the graph.  
> geeks<-c(0.5,1.5,3.5,5.4)  
> labels<-c("2000","2005","2010","2015")  
> piepercent<-round(100*geeks/sum(geeks),1)  
> #Plot the chart.  
> pie(geeks,labels=piepercent,main="Electricity Production from Renewable Sources - INDIA",col=rainbow(length(geeks)))  
> legend("topright",c("2000","2005","2010","2015"),  
+       cex=0.5,fill=rainbow(length(geeks)))  
>  
> |
```

CHINA

The RStudio interface displays the following R code in the editor:

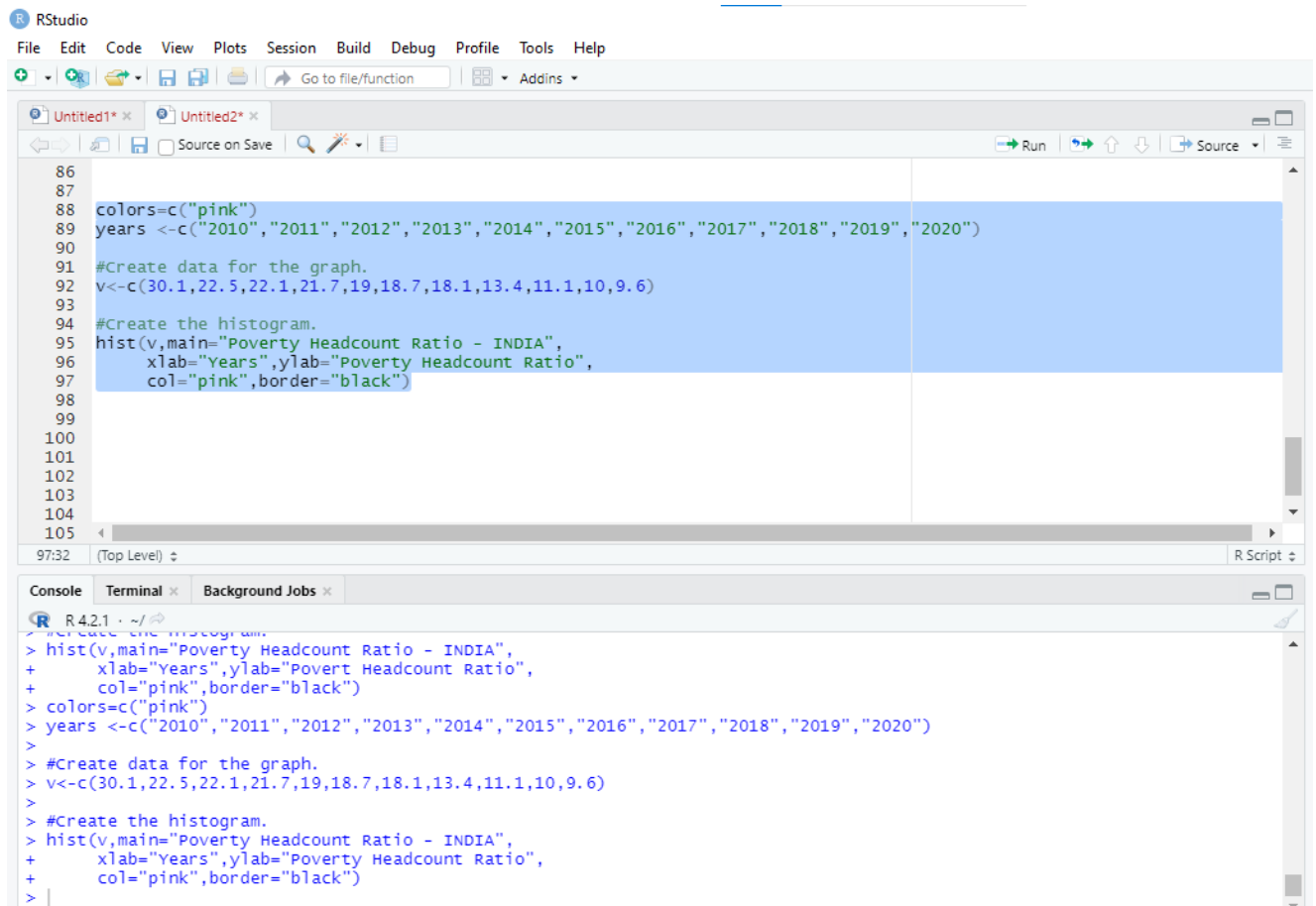
```
44  
45 #Create data for the graph.  
46 geeks<-c(0.2,0.3,1.7,4.9)  
47 labels<-c("2000","2005","2010","2015")  
48  
49 piepercent<-round(100*geeks/sum(geeks),1)  
50  
51 #Plot the chart.  
52 pie(geeks,labels=piepercent,main="Electricity Production from Renewable Sources - CHINA",col=rainbow(length(geeks)))  
53 legend("topright",c("2000","2005","2010","2015"),  
54       cex=0.5,fill=rainbow(length(geeks)))  
55  
56  
57  
58  
59  
60  
61  
62  
63
```

The console shows the execution of the code:

```
> #Create data for the graph.  
> geeks<-c(0.2,0.3,1.7,4.9)  
> labels<-c("2000","2005","2010","2015")  
> piepercent<-round(100*geeks/sum(geeks),1)  
> #Plot the chart.  
> pie(geeks,labels=piepercent,main="Electricity Production from Renewable Sources - CHINA",col=rainbow(length(geeks)))  
> legend("topright",c("2000","2005","2010","2015"),  
+       cex=0.5,fill=rainbow(length(geeks)))  
>  
>
```

- **POVERTY HEADCOUNT RATIO 2010-2020**
REPRESENTED BY HISTOGAM

INDIA



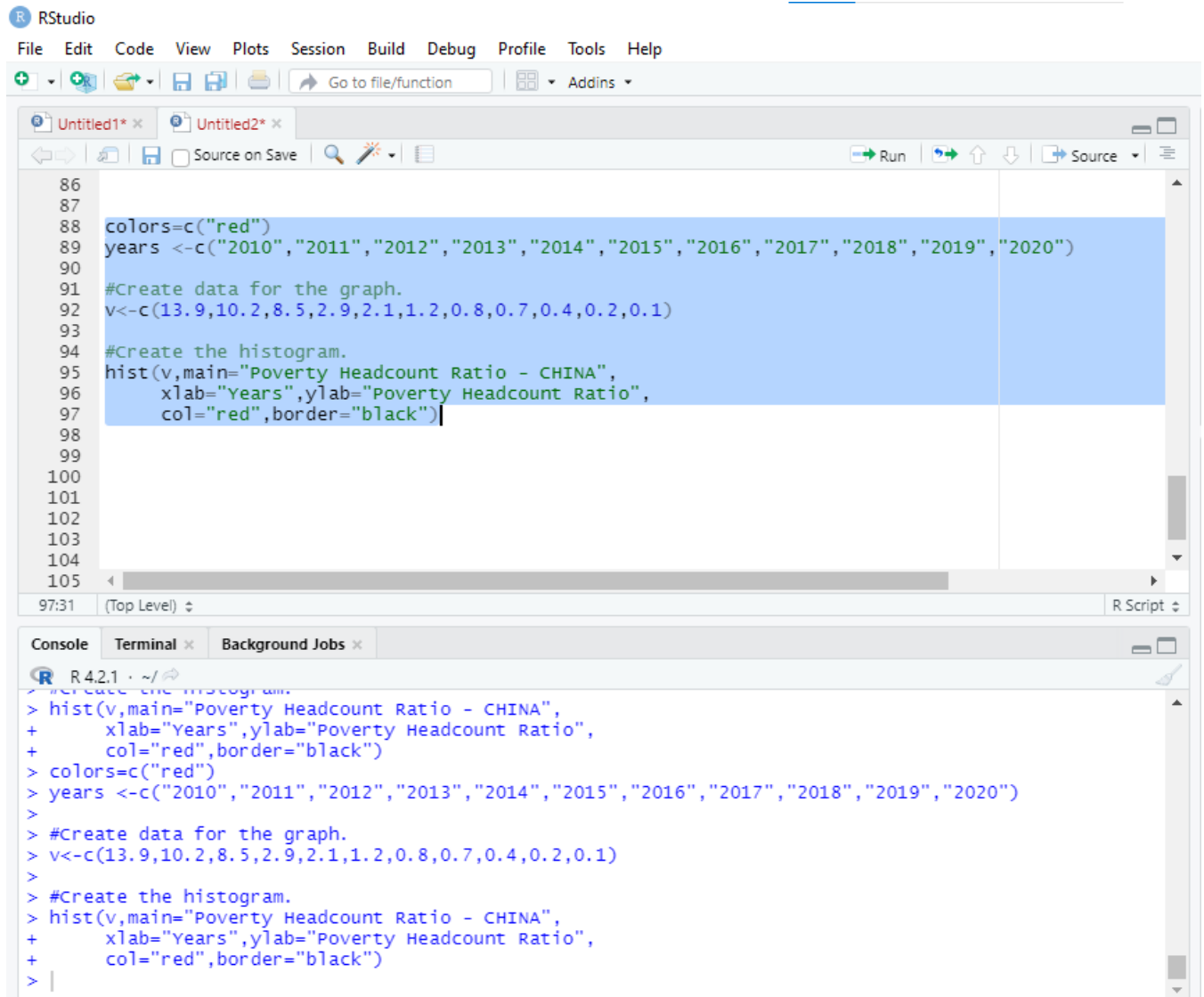
The screenshot displays the RStudio interface with a script editor and a console. The script editor contains R code for creating a histogram. The console shows the execution of the code, with some lines being repeated.

```
86
87
88 colors=c("pink")
89 years <-c("2010","2011","2012","2013","2014","2015","2016","2017","2018","2019","2020")
90
91 #Create data for the graph.
92 v<-c(30.1,22.5,22.1,21.7,19,18.7,18.1,13.4,11.1,10,9.6)
93
94 #Create the histogram.
95 hist(v,main="Poverty Headcount Ratio - INDIA",
96     xlab="Years",ylab="Poverty Headcount Ratio",
97     col="pink",border="black")
98
99
100
101
102
103
104
105
```

Console output:

```
R 4.2.1 ~ /
> #Create the histogram.
> hist(v,main="Poverty Headcount Ratio - INDIA",
+     xlab="Years",ylab="Povert Headcount Ratio",
+     col="pink",border="black")
> colors=c("pink")
> years <-c("2010","2011","2012","2013","2014","2015","2016","2017","2018","2019","2020")
>
> #Create data for the graph.
> v<-c(30.1,22.5,22.1,21.7,19,18.7,18.1,13.4,11.1,10,9.6)
>
> #Create the histogram.
> hist(v,main="Poverty Headcount Ratio - INDIA",
+     xlab="Years",ylab="Poverty Headcount Ratio",
+     col="pink",border="black")
>
```

CHINA



The screenshot displays the RStudio interface with two panes. The top pane shows the source editor with R code for creating a histogram. The bottom pane shows the console with the executed commands.

```
86
87
88 colors=c("red")
89 years <-c("2010","2011","2012","2013","2014","2015","2016","2017","2018","2019","2020")
90
91 #Create data for the graph.
92 v<-c(13.9,10.2,8.5,2.9,2.1,1.2,0.8,0.7,0.4,0.2,0.1)
93
94 #Create the histogram.
95 hist(v,main="Poverty Headcount Ratio - CHINA",
96      xlab="Years",ylab="Poverty Headcount Ratio",
97      col="red",border="black")
98
99
100
101
102
103
104
105
```

Console output:

```
R 4.2.1 ~ /
> #Create the histogram.
> hist(v,main="Poverty Headcount Ratio - CHINA",
+      xlab="Years",ylab="Poverty Headcount Ratio",
+      col="red",border="black")
> colors=c("red")
> years <-c("2010","2011","2012","2013","2014","2015","2016","2017","2018","2019","2020")
>
> #Create data for the graph.
> v<-c(13.9,10.2,8.5,2.9,2.1,1.2,0.8,0.7,0.4,0.2,0.1)
>
> #Create the histogram.
> hist(v,main="Poverty Headcount Ratio - CHINA",
+      xlab="Years",ylab="Poverty Headcount Ratio",
+      col="red",border="black")
>
```

- **LIFE EXPECTANCY AT BIRTH 2010-2020 PRESENTED BY BOX PLOT**

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

Filter

	YEAR	INDIA	CHINA
1	2010	67.1	74.0
2	2011	67.5	75.2
3	2012	68.3	75.5
4	2013	68.7	75.9
5	2014	69.0	76.1
6	2015	69.2	76.4
7	2016	69.6	76.5
8	2017	69.7	76.8
9	2018	69.9	77.3
10	2019	70.1	77.7
11	2020	70.4	77.9

Showing 1 to 11 of 11 entries, 3 total columns

Console Terminal Background Jobs

```
R 4.2.1 ~
> boxplot(Bangladesh,India,col="green",main="Boxplot of Life Expectancy at Birth",xlab="years",ylab="Life Expectancy")
> library(readxl)
> ABCD <- read_excel("ABCD.xlsx")
> view(ABCD)
> boxplot(china,India,col="violet",main="Boxplot of Life Expectancy at Birth",ylab="Age(Years)")
Error in boxplot(china, India, col = "violet", main = "Boxplot of Life Expectancy at Birth", :
  object 'china' not found
> boxplot(china,India,col="violet",main="Boxplot of Life Expectancy at Birth",ylab="Age Years")
```

- **GDP REVIEW ON THE BASIS OF DIFFERENT STATISTICAL TOOLS**

INDIA

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

```
122 dataset<-c(1.68,1.82,1.83,1.86,2.04,2.10,2.29,2.65,2.70,2.83,2.67,3.17)
123
124 mean(x=dataset)
125 median(x=dataset)
126 mode(x=dataset)
127 mean(x=dataset,trim=0.1)
128 length(x=dataset)
129 var(x=dataset)
130 sd(x=dataset)
131 min(x=dataset)
132 max(x=dataset)
133 range(x=dataset)
134 IQR(x=dataset)
135 summary(x=dataset)
136
137
138
139
140
```

135:19 (Top Level) R Script

Console Terminal Background Jobs

```
R 4.2.1 ~ /
> dataset<-c(1.68,1.82,1.83,1.86,2.04,2.10,2.29,2.65,2.70,2.83,2.67,3.17)
> mean(x=dataset)
[1] 2.303333
> median(x=dataset)
[1] 2.195
> mode(x=dataset)
[1] "numeric"
> mean(x=dataset,trim=0.1)
[1] 2.279
> length(x=dataset)
[1] 12
> var(x=dataset)
[1] 0.2358242
> sd(x=dataset)
[1] 0.4856174
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

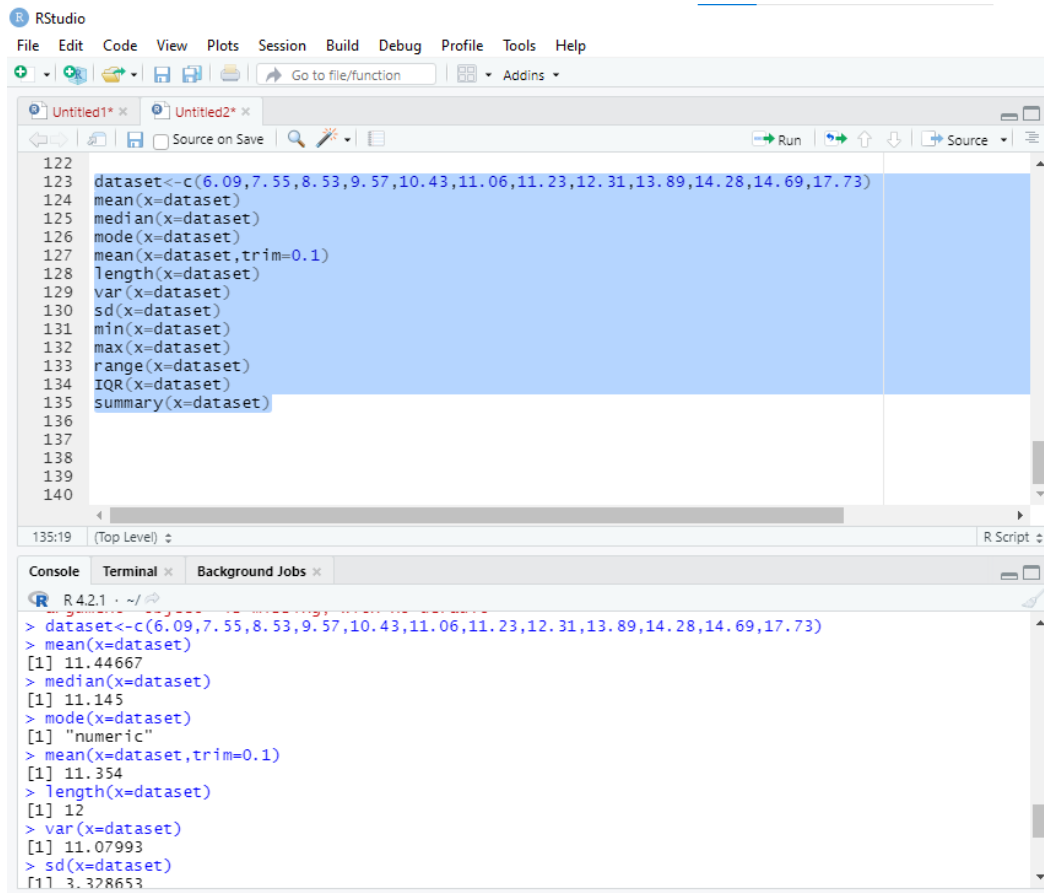
```
122 dataset<-c(1.68,1.82,1.83,1.86,2.04,2.10,2.29,2.65,2.70,2.83,2.67,3.17)
123
124 mean(x=dataset)
125 median(x=dataset)
126 mode(x=dataset)
127 mean(x=dataset,trim=0.1)
128 length(x=dataset)
129 var(x=dataset)
130 sd(x=dataset)
131 min(x=dataset)
132 max(x=dataset)
133 range(x=dataset)
134 IQR(x=dataset)
135 summary(x=dataset)
136
137
138
139
140
```

135:19 (Top Level) R Script

Console Terminal Background Jobs

```
R 4.2.1 ~ /
> length(x=dataset)
[1] 12
> var(x=dataset)
[1] 0.2358242
> sd(x=dataset)
[1] 0.4856174
> min(x=dataset)
[1] 1.68
> max(x=dataset)
[1] 3.17
> range(x=dataset)
[1] 1.68 3.17
> IQR(x=dataset)
[1] 0.825
> summary(x=dataset)
```

CHINA

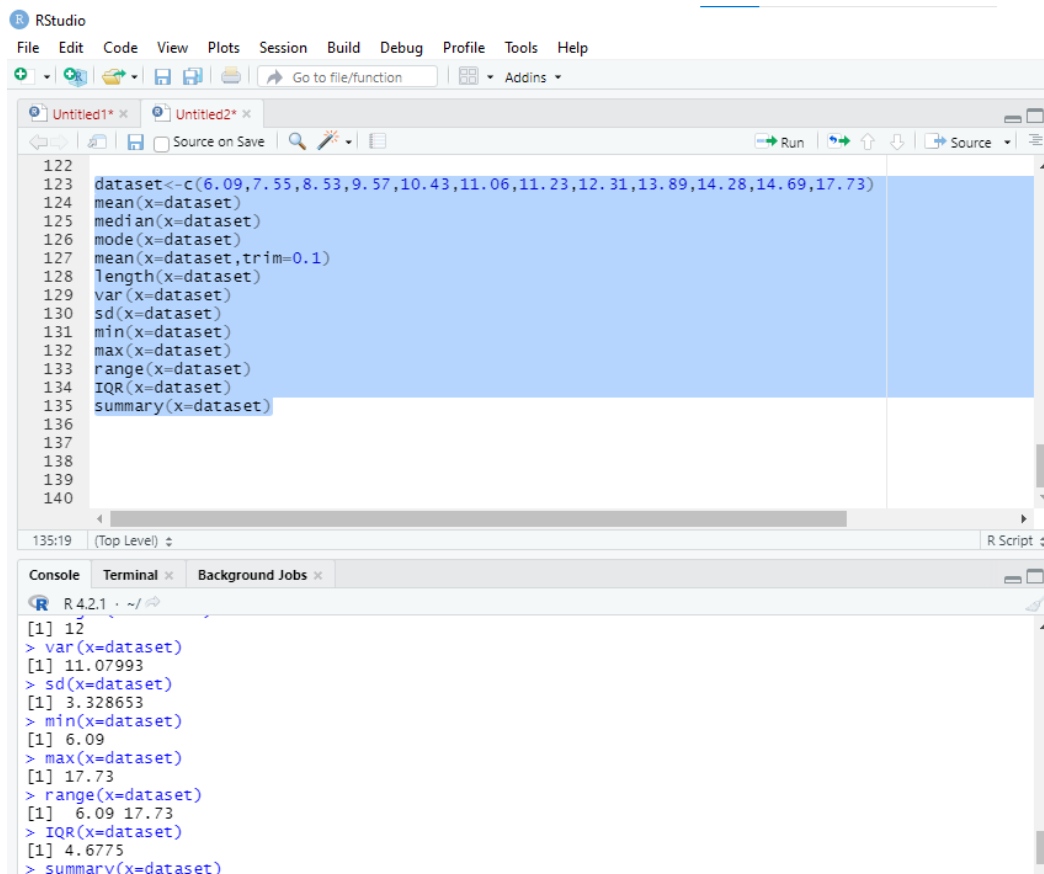


The screenshot shows the RStudio interface with a script editor and a console. The script editor contains the following code:

```
122 dataset<-c(6.09,7.55,8.53,9.57,10.43,11.06,11.23,12.31,13.89,14.28,14.69,17.73)
123 mean(x=dataset)
124 median(x=dataset)
125 mode(x=dataset)
126 mean(x=dataset,trim=0.1)
127 length(x=dataset)
128 var(x=dataset)
129 sd(x=dataset)
130 min(x=dataset)
131 max(x=dataset)
132 range(x=dataset)
133 IQR(x=dataset)
134 summary(x=dataset)
```

The console shows the output of the script:

```
R 4.2.1 ~ /
> dataset<-c(6.09,7.55,8.53,9.57,10.43,11.06,11.23,12.31,13.89,14.28,14.69,17.73)
> mean(x=dataset)
[1] 11.44667
> median(x=dataset)
[1] 11.145
> mode(x=dataset)
[1] "numeric"
> mean(x=dataset,trim=0.1)
[1] 11.354
> length(x=dataset)
[1] 12
> var(x=dataset)
[1] 11.07993
> sd(x=dataset)
[1] 3.328653
```



The screenshot shows the RStudio interface with a script editor and a console. The script editor contains the following code:

```
122 dataset<-c(6.09,7.55,8.53,9.57,10.43,11.06,11.23,12.31,13.89,14.28,14.69,17.73)
123 mean(x=dataset)
124 median(x=dataset)
125 mode(x=dataset)
126 mean(x=dataset,trim=0.1)
127 length(x=dataset)
128 var(x=dataset)
129 sd(x=dataset)
130 min(x=dataset)
131 max(x=dataset)
132 range(x=dataset)
133 IQR(x=dataset)
134 summary(x=dataset)
```

The console shows the output of the script:

```
R 4.2.1 ~ /
[1] 12
> var(x=dataset)
[1] 11.07993
> sd(x=dataset)
[1] 3.328653
> min(x=dataset)
[1] 6.09
> max(x=dataset)
[1] 17.73
> range(x=dataset)
[1] 6.09 17.73
> IQR(x=dataset)
[1] 4.675
> summary(x=dataset)
```